

IIDS67692 Computational Methods for Multimodal Data Analysis

Deadline for submission: **02/06/2026 at 2pm** (or one week later for those with an extension)
See the 'Submission guidance' section for submission details.

Instructions - Please read carefully

The code and datasets for this assignment are available on Canvas and can also be found in the Nuvolos **"Assignment"** folder.

You can run the notebooks in Nuvolos or in any environment of your choice, given adequate computational resources.

In Nuvolos, each student will be allocated **1 credit quota**, which corresponds to approximately **7 hours of GPU usage; you might also have left over credits from your lab sessions.**

Please note that **GPU credits are consumed for the entire time the GPU notebook is active**, not just when your model is training. This means that simply having the GPU notebook open will consume your credits. Don't forget to stop the notebooks when inactive.

Being mindful of your GPU usage will help you avoid running out of credits prematurely.

PART A

In the **Assignment_PartA.ipynb** file you will find code to load in a publicly available spatial transcriptomics 10x Visum dataset from the [dorsolateral prefrontal cortex \(DLPFC\)](#) from postmortem human brain samples (folder name 151676). You can also download the dataset and the notebook from Canvas ([151676.zip](#)).

The folder contains the following:

- **filtered_feature_bc_matrix.h5** : This is the main **gene expression** matrix which stores gene expression counts information as well as spatial positions.
- **Folder: spatial/**:
 - This folder contains all the **spatial imaging** and **coordinate information** needed to map gene expression back onto the tissue image.
 - **Histology images**: Usually: **tissue_hires_image.png** and **tissue_lowres_image.png**. Our folder also contains also a full resolution histology image: **151676_full_image.tif**. **You will use this image to extract features.**
- **151676_truth.txt** This file contains ground-truth annotations for tissue domains.

The task is to use SpaGCN to identify and cluster tissue domains (cortical layers). You will then evaluate clustering performance against manually annotated cortical layer labels (provided ground-truth) and investigate how the contribution of histology information affects the results.

TASKS

1. Discuss the computational method that SpaGCN is based on. Which modalities are integrated and what type of fusion is it using?
2. Load the Visium dataset and the histology image (151676_full_image.tif)
3. Preprocess the Data
4. Use the SpaGCN pipeline with image parameter $s=1$ to identify spatial tissue domains
5. Cluster the domains using **k-means** with number of clusters equal to the number of ground-truth labels
6. Visualise the clusters spatially
7. Evaluate the clustering performance against the ground-truth labels
8. Assess how well each tissue domain/layer has been recovered
9. Run SpaGCN again using the following values of the histology contribution parameter and **compare with the $s=1$:**
 - **$s=0$**
 - **$s=10$**
10. Discuss how the histology contribution parameter affects clustering performance on average and per label

For PART A you must complete all tasks above using either the **LAB Week 4 Python 3.8 Jupyter notebook** in Nuvolos or any environment of your choice, given adequate computational resources.

PART B

In the Assignment_PartB.ipynb file, you will find code to build a multimodal LLM for a **health decision support** via a **visual question answering (VQA) task**. The code includes a dataloader that uses a subset of the publicly available PathVQA dataset from the Hugging Face dataset library, consisting of pathology images and their corresponding question-answer pairs for clinical decision support.

For this application please use the [LLM-GPU] JUPYTERLAB 4.5.4 WITH PYTHON 3.13 AND JULIA 1.12.4 notebook in Nuvolos or any environment of your choice, given adequate computational resources.

The task is to build a multimodal LLM with a gated cross-attention fusion technique to adapt the GPT-2 model for the pathology visual question answering task.

TASKS

1. Discuss the PathVQA health decision support dataset, including visualising 3 samples of images, questions, and corresponding answers. The samples can be:
 - a. (i) sample with minimum QA length
 - b. (ii) sample with average QA length
 - c. (iii) sample with maximum QA length
2. Use "Cross-Attention Fusion" and complete the code for LoRA training, inference (for few-sample QA and prediction visualisation), and inference for evaluation metrics. Train the model and run inference to collect the prediction visualisations and evaluation metrics (BLEU, ROUGE-L, and METEOR). Create Fig. 1 for the prediction visualisations and Table 1 for the metrics.
3. Design a "Gated Cross-Attention Fusion" technique and replace the previous "Cross-Attention Fusion" method. Observe the performance changes and record the prediction visualisations and metrics in a new row of Fig. 1 and Table 1.
4. Discuss the performance of "Gated Cross-Attention Fusion" vs. "Cross-Attention Fusion" and explain the reasons for any performance gain or drop with "Gated Cross-Attention Fusion".
5. Conduct hyperparameter tuning (e.g., learning rate, epochs) and observe the performance changes in Table 1 and Fig. 1. Discuss the performance differences before and after tuning.
 - why? because for number of dataset or what...
 - 1. learning rate
 - 2. epoch
 - 3. Lora change to 5, 6, 7, 8, 9 -> depend by data

Submission guidance

- Your submission should include the following three parts and must be submitted to **Canvas** by the advertised deadline.
- **Report:** The 1000-word count should be distributed equally between Parts A and B. The word count excludes figure/table legends and the bibliography. No abstract or introduction is required.
- **Two Jupyter notebooks:** You should complete and run the notebooks in Nuvolos or in any other environment of your choice, provided adequate computational resources are available. Once completed, download the notebooks and upload them to Canvas. Naming convention: Name your Jupyter notebooks **your_ID_AssessmentPartA.ipynb** and **your_ID_AssessmentPartB.ipynb**.

Marking scale:

Methods and Results: 60 %

Discussion: 10%

Presentation: 10%

Code: 20 %

